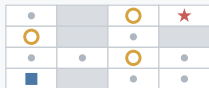


a Primitive finite MDP



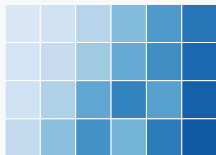
many primitive states $|S|$

■ ★ mandatory boundary

○ turn / articulation
candidate universe

b Frozen Green response

$$H_K = \sum_{t=0}^K (\gamma P_{II})^t P_{IC}$$



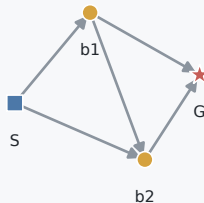
schematic first-hit exposure

● flow ● Green ● control

● topology ● stochastic
one sparse message-passing transform
no candidate insertion or recompute

c Threshold once

local maxima with $q \geq 2$ support
topology or stochastic gate



selected boundary graph

build kernels once
option edges replace interior states

d Audit and plan



pass
graph-SMDP
planning

fail
adaptive RD
fallback

certify approximation error
not global RD optimality